

METEOR TRAIL

Meteors are randomly occurring, narrow streaks of light that shoot across a few tens of degrees of the sky in just a few seconds. A typical meteor will be about 90 miles (150 km) away from the observer.

METEORITE IMPACTS

Earth's atmosphere shields the surface from the vast majority of incoming extraterrestrial bodies. The typical impact velocity at the top of the atmosphere is about 45,000 mph (72,000 kph), and the leading surface of the meteoroid quickly heats up and starts boiling as a result of hitting air molecules at this speed. Usually, the body is so small that it boils away completely. Parts of medium-sized bodies survive to fall as meteorites. However, a very large body, having a mass greater than about 100,000 tons, is hardly affected by the atmosphere. It punches through the gas like a bullet through tissue paper, energetically slamming into Earth's surface and gouging out a circular crater that is typically 20 times larger than its own size (see p.103). The enormous energy generated ensures that most of the impactor is vaporized in the process, and seismic shocks and blast waves are produced. The resulting huge earthquake will topple any trees for many miles around. The surrounding atmosphere reaches furnace temperatures, causing widespread fires. A tsunami will be produced if the impact is in the ocean.

**IMPACT CRATER**

About 50,000 years ago, an iron meteorite hit this desert region in Arizona. The resulting crater, called Meteor Crater, is 0.75 miles (1.2 km) wide and 550 ft (170 m) deep. Ejecta produced by the impact can be seen as hummocky deposits lying beyond the crater rim.

An impact crater greater than 12 miles (20 km) in diameter is produced on Earth about once every 500,000 years.

**MOLDAVITE (GREEN GLASS)****DISK-SHAPED TEKTITE****IMPACTITES**

These half-inch, glassy bodies are formed when Earth's rock melts or shatters due to the heat and pressure of an impact.

EXPLORING SPACE**FINDING METEORITES**

The best way to find meteorites is to search exposed glacial ice fields or sandy deserts free of other large rock. Ideal sites are the eroded blue-ice regions of Antarctica and the Nullarbor Plain in Australia. Since 1976, US, European, and Japanese expeditions have searched for meteorites in Antarctica, and thousands of individual specimens have been recovered. Many are from the same fall, due to the incoming body fragmenting as it passed through Earth's atmosphere.

**NOMAD ROVER IN ANTARCTICA**

In January 2000, the US robot Nomad achieved a first by finding and identifying five meteorites lying on the ice in eastern Antarctica using just sensors and artificial intelligence.